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(54) **DISPLAY APPARATUS FOR VEHICLE**

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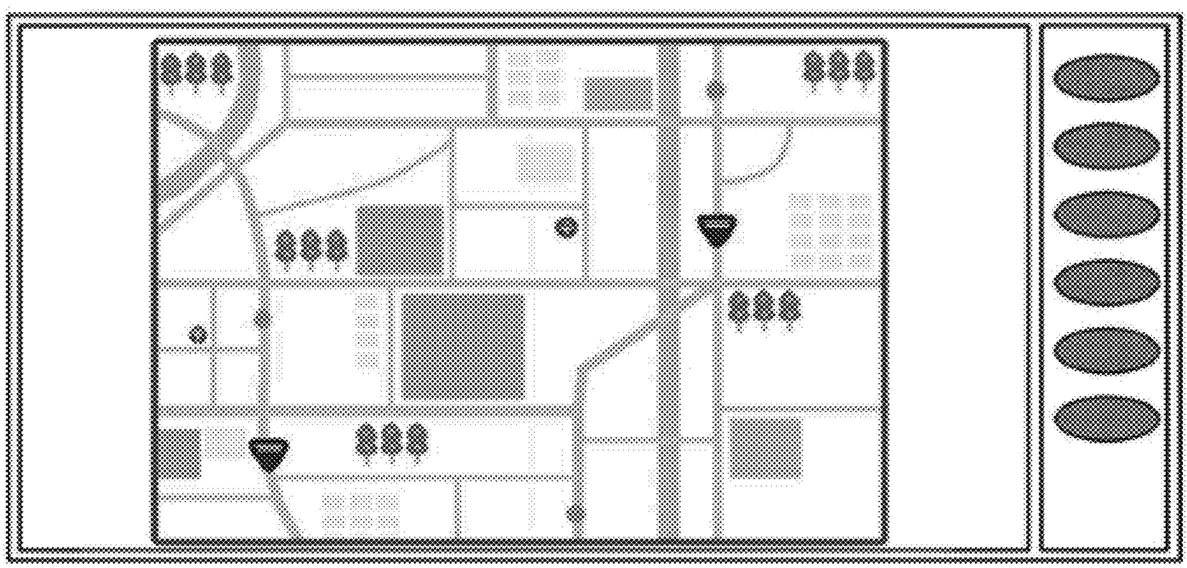
(57) **ABSTRACT**

A vehicle display apparatus includes: a display including a first region in which provided content is to be displayed and a second region in which a link icon linking to the content is to be displayed; a touch sensor; a display control unit controlling a display mode of the display; and a storage holding data on the display mode. The display control unit keeps a display area of the second region smaller than that of the first region, and expands, when the touch sensor detects a swiping operation made by an occupant in the vehicle on the link icon, a rendering region of the second region, based on a completion position of the swiping operation, within a range where the display area of the second region is kept smaller than that of the first region, while continuing to execute display control in the first region.



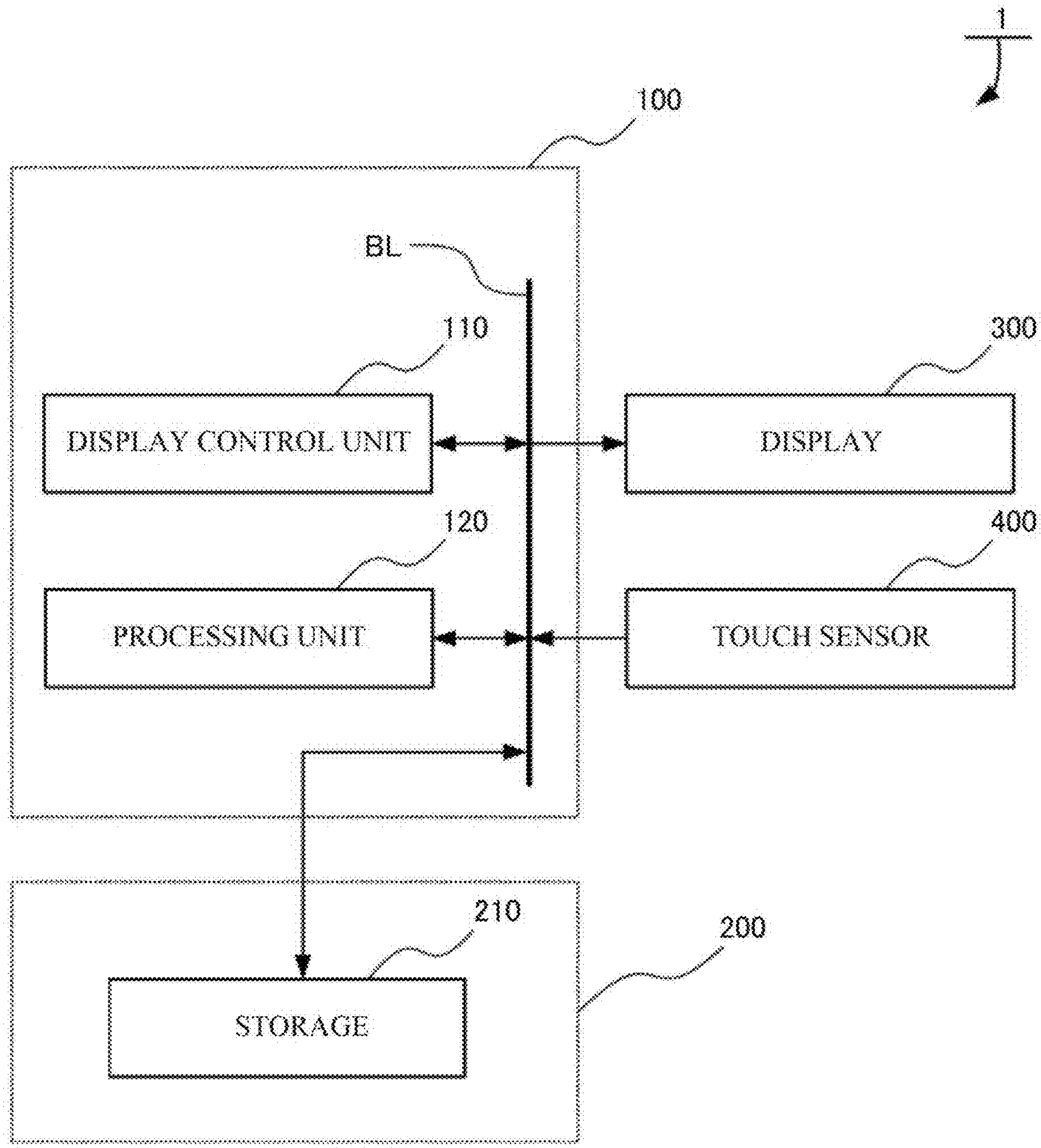

1
100
BL
110
DISPLAY CONTROL UNIT
300
DISPLAY
120
PROCESSING UNIT
400
TOUCH SENSOR
210
200
STORAGE

FIG. 1

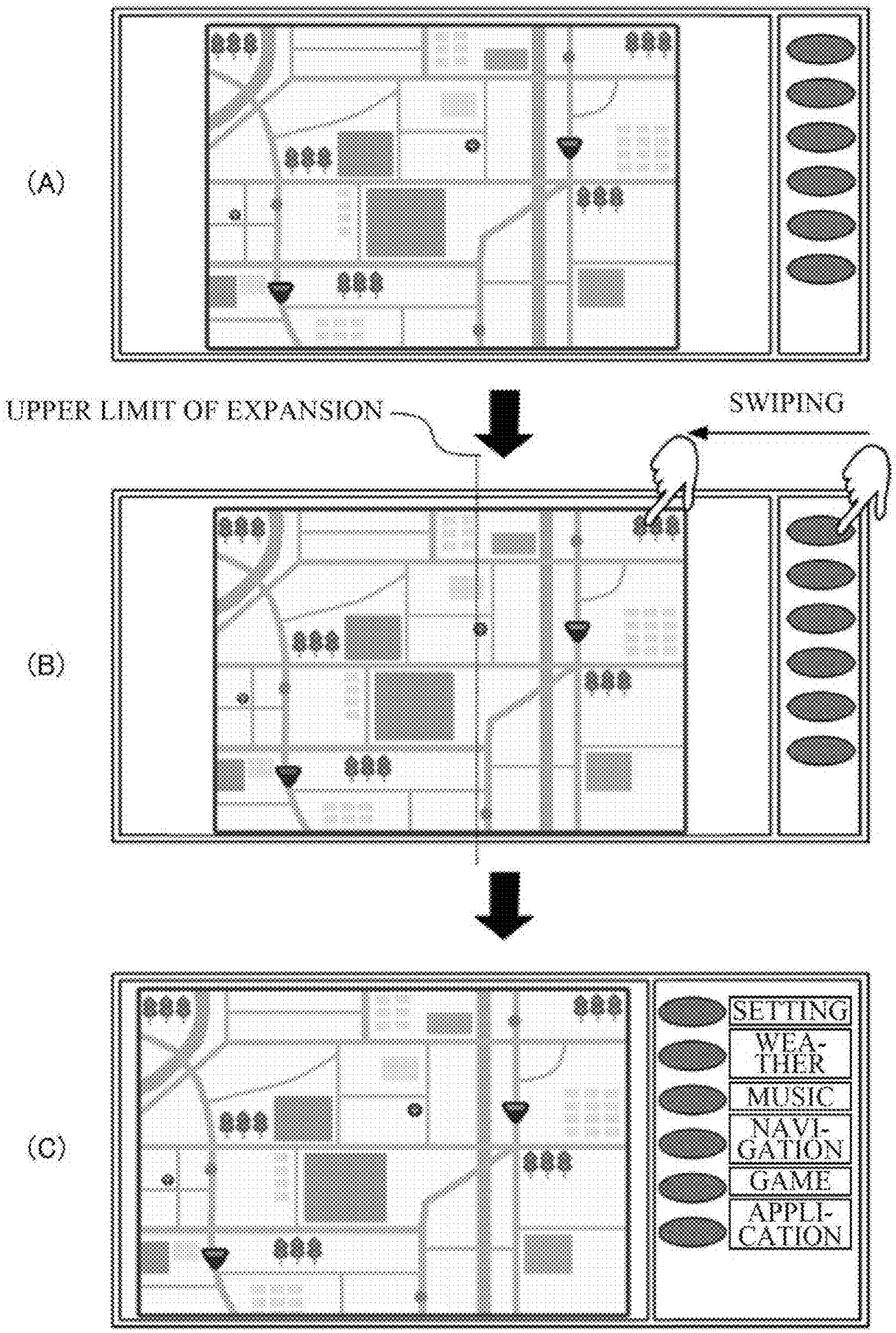

(A)
UPPER LIMIT OF EXPANSION
SWIPING
(B)
(C)
SETTING
WEA-THER
MUSIC
NAVI-GATION
GAME
APPLI-CATION

FIG. 2

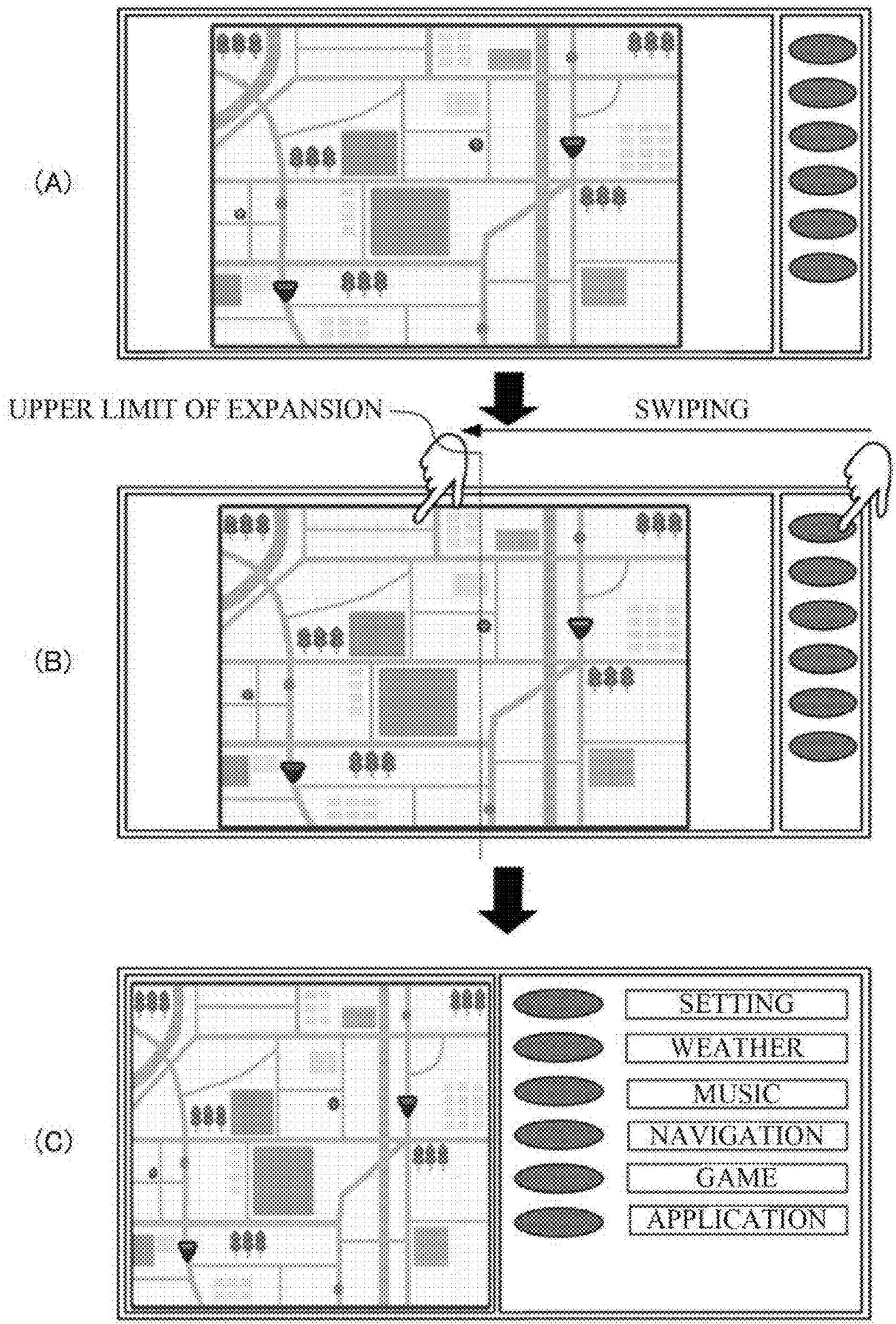

(A)
UPPER LIMIT OF EXPANSION
SWIPING
(B)
(C)
SETTING
WEATHER
MUSIC
NAVIGATION
GAME
APPLICATION

FIG. 3

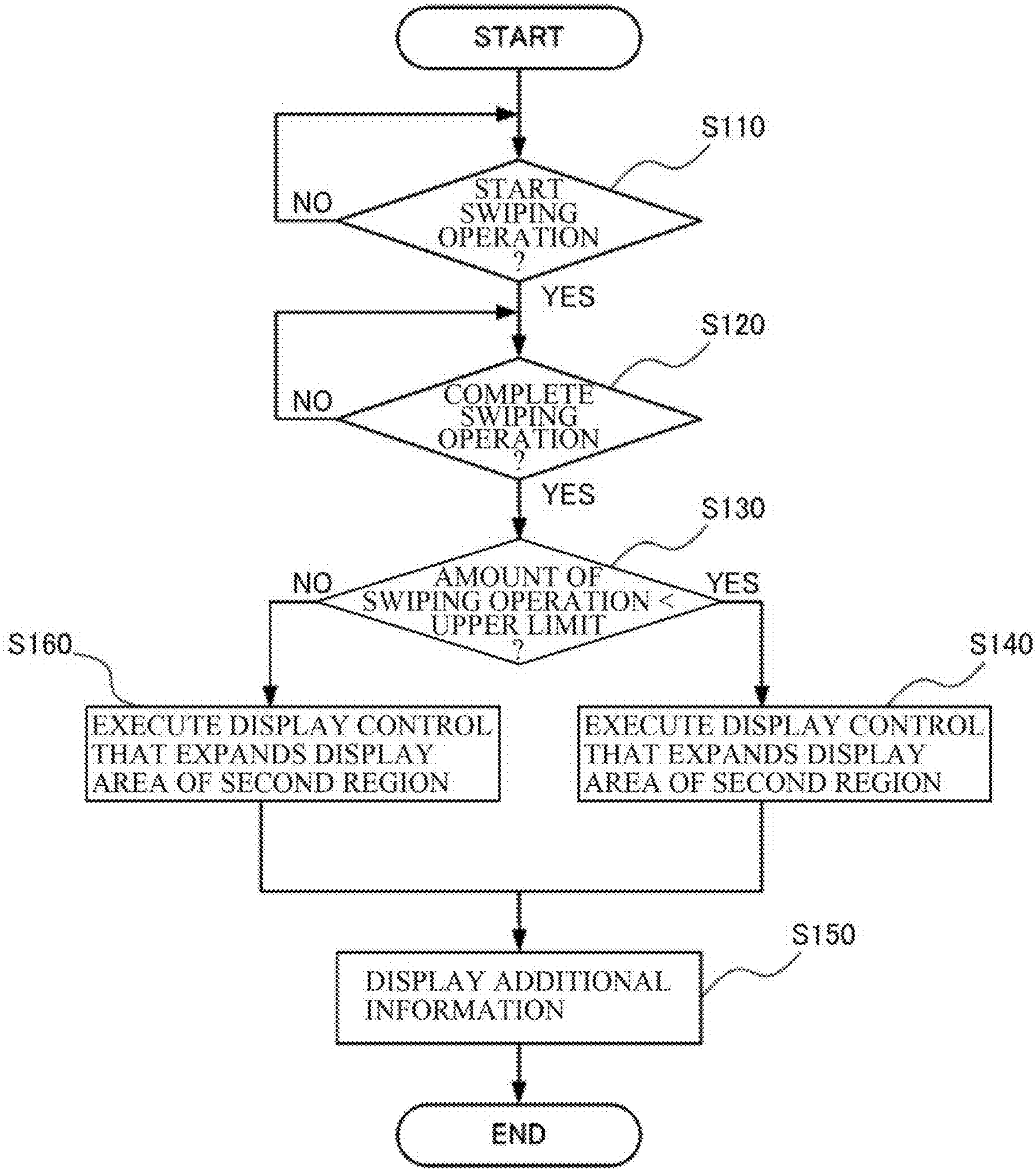

START
S110
START SWIPING OPERATION ?
NO
YES
S120
COMPLETE SWIPING OPERATION ?
NO
YES
S130
AMOUNT OF SWIPING OPERATION < UPPER LIMIT ?
NO
YES
S160
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S140
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S150
DISPLAY ADDITIONAL INFORMATION
END

FIG. 4

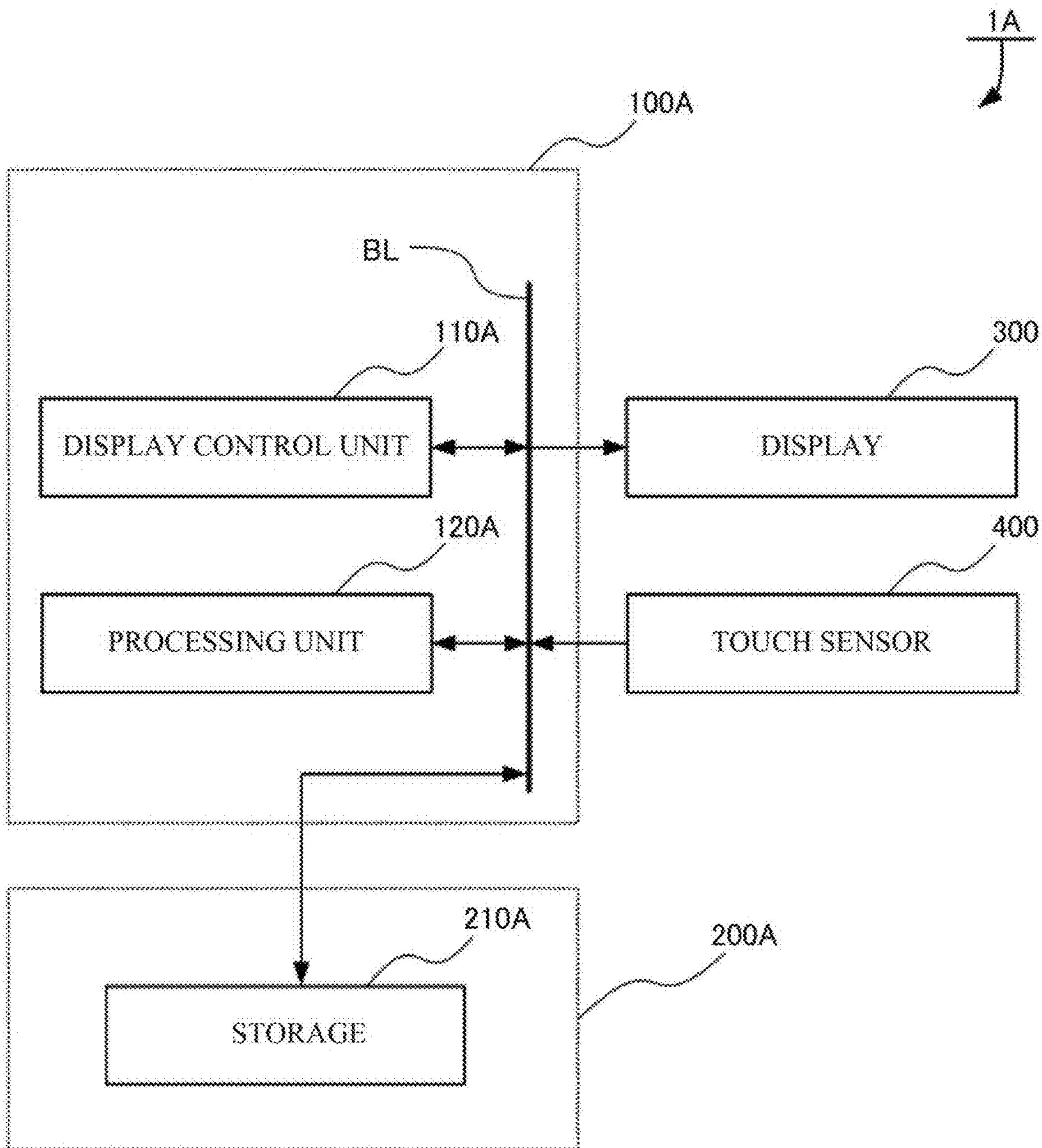

1A
100A
BL
110A
DISPLAY CONTROL UNIT
300
DISPLAY
120A
PROCESSING UNIT
400
TOUCH SENSOR
210A
200A
STORAGE

FIG. 5

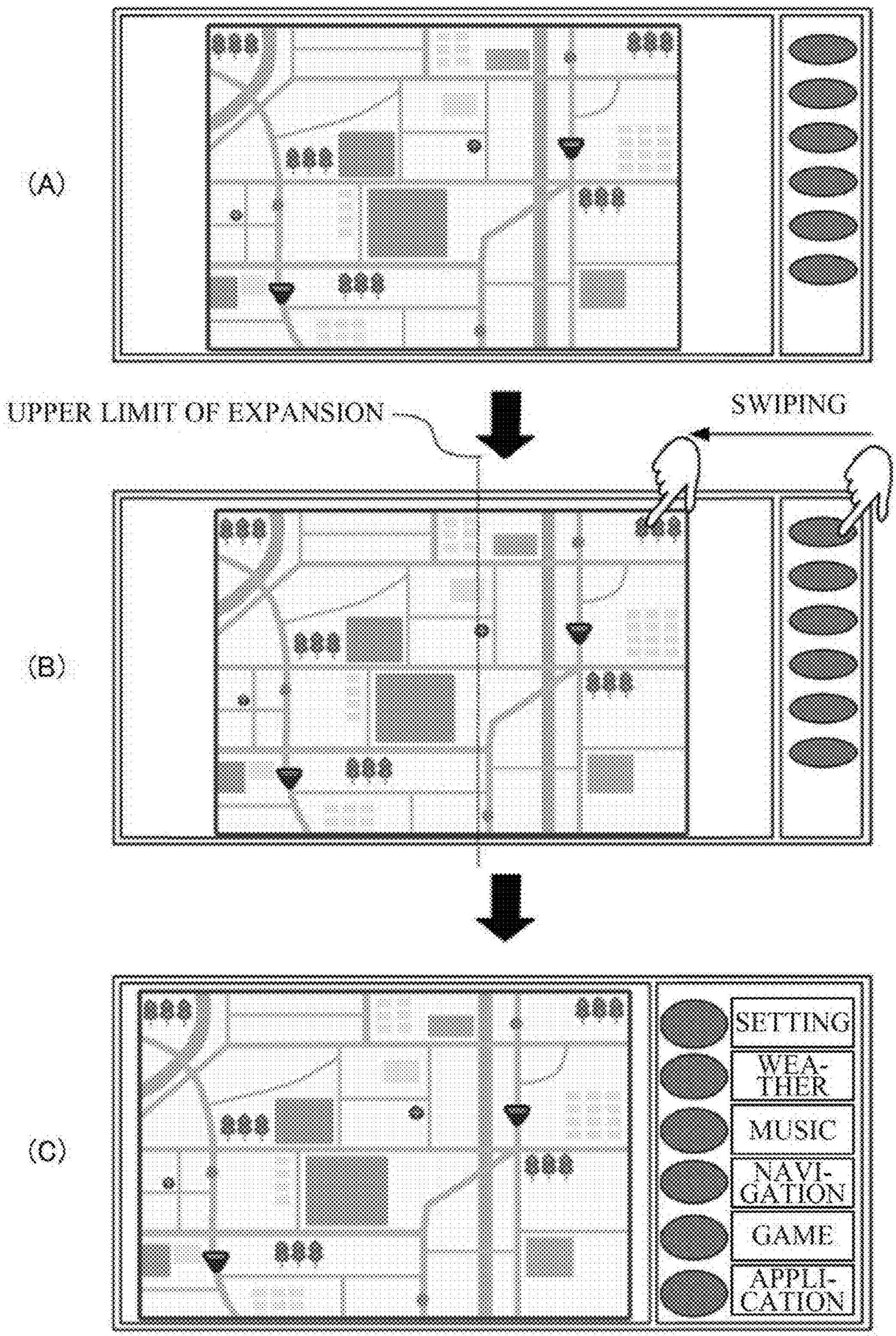

(A)
UPPER LIMIT OF EXPANSION
SWIPING
(B)
(C)
SETTING
WEA-
THER
MUSIC
NAVI-
GATION
GAME
APPLI-
CATION

FIG. 6

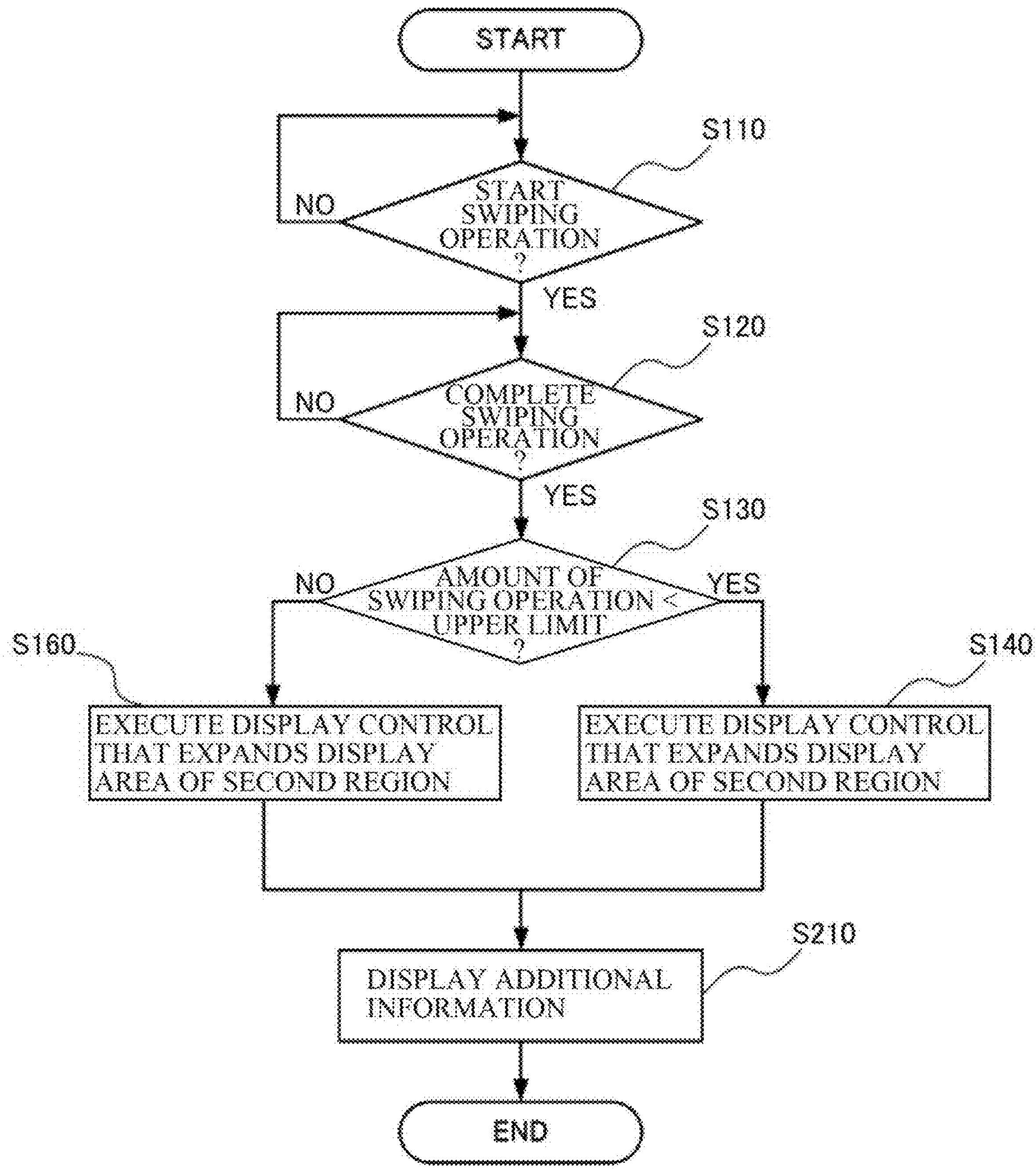

START
S110
START SWIPING OPERATION ?
NO
YES
S120
COMPLETE SWIPING OPERATION ?
NO
YES
S130
AMOUNT OF SWIPING OPERATION < UPPER LIMIT ?
NO
YES
S160
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S140
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S210
DISPLAY ADDITIONAL INFORMATION
END

FIG. 7

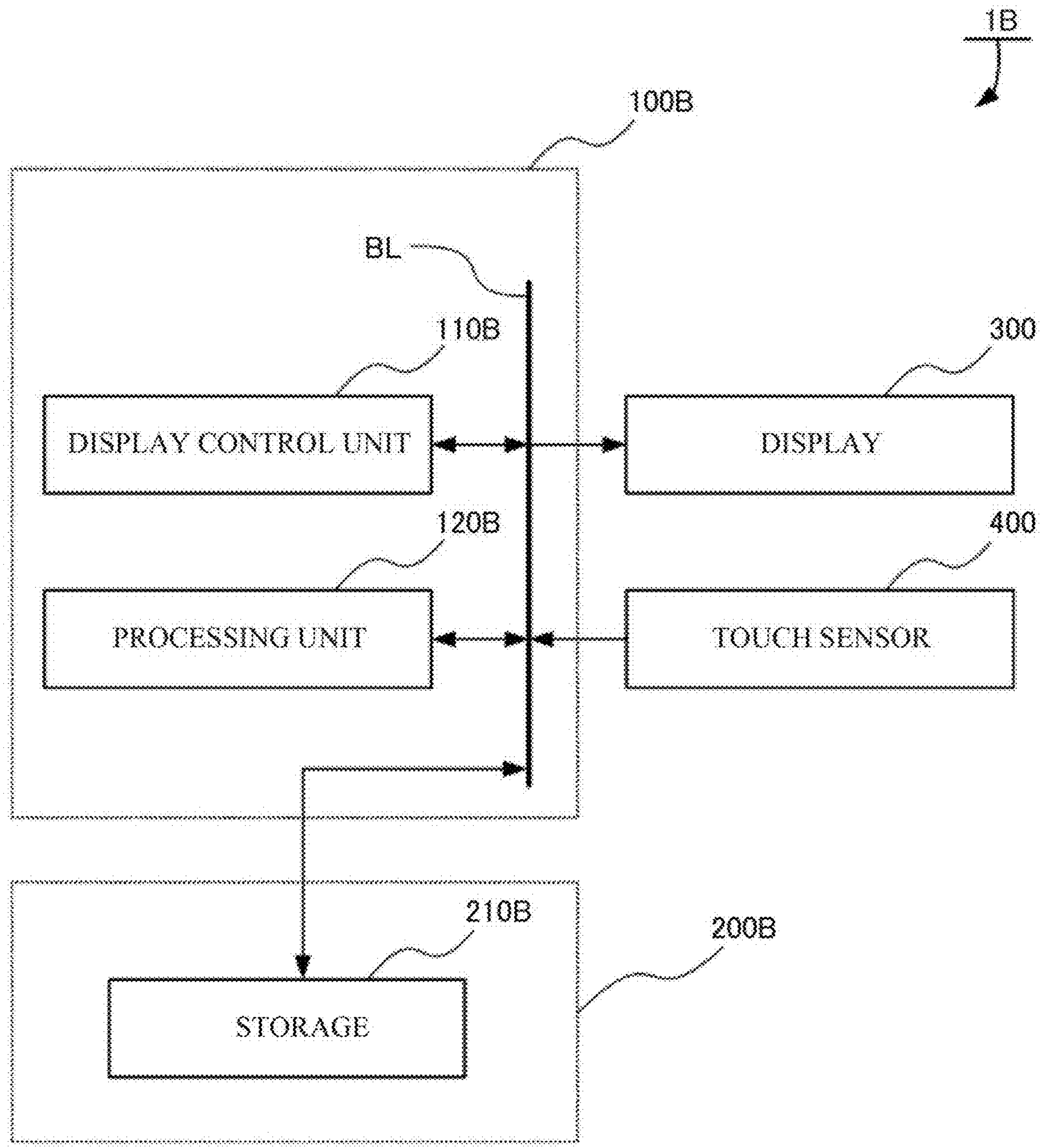

1B
100B
BL
110B
DISPLAY CONTROL UNIT
300
DISPLAY
120B
PROCESSING UNIT
400
TOUCH SENSOR
210B
STORAGE
200B

FIG. 8

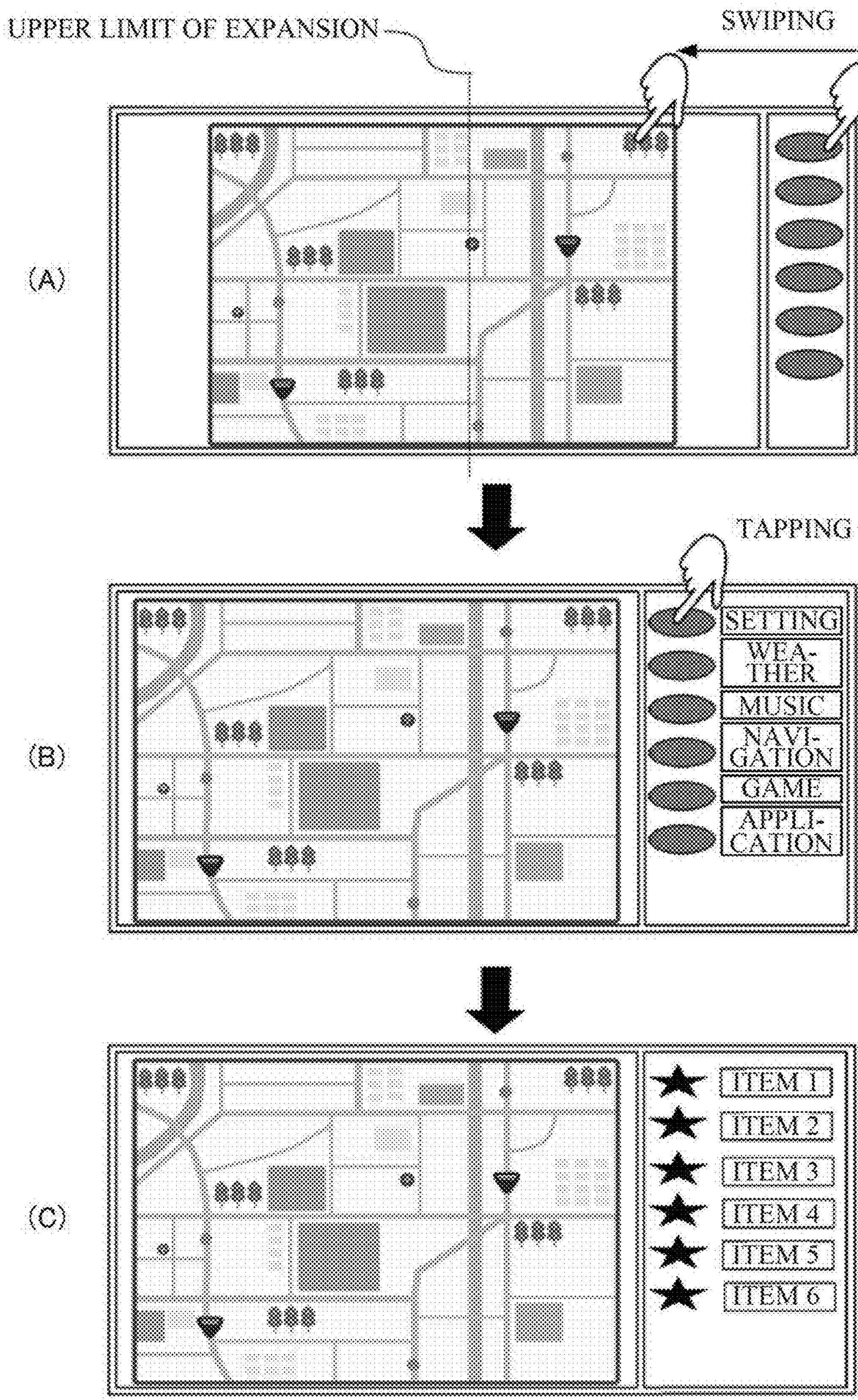

UPPER LIMIT OF EXPANSION
SWIPING
(A)
TAPPING
(B)
SETTING
WEA-
THER
MUSIC
NAVI-
GATION
GAME
APPLI-
CATION
(C)
ITEM 1
ITEM 2
ITEM 3
ITEM 4
ITEM 5
ITEM 6

FIG. 9

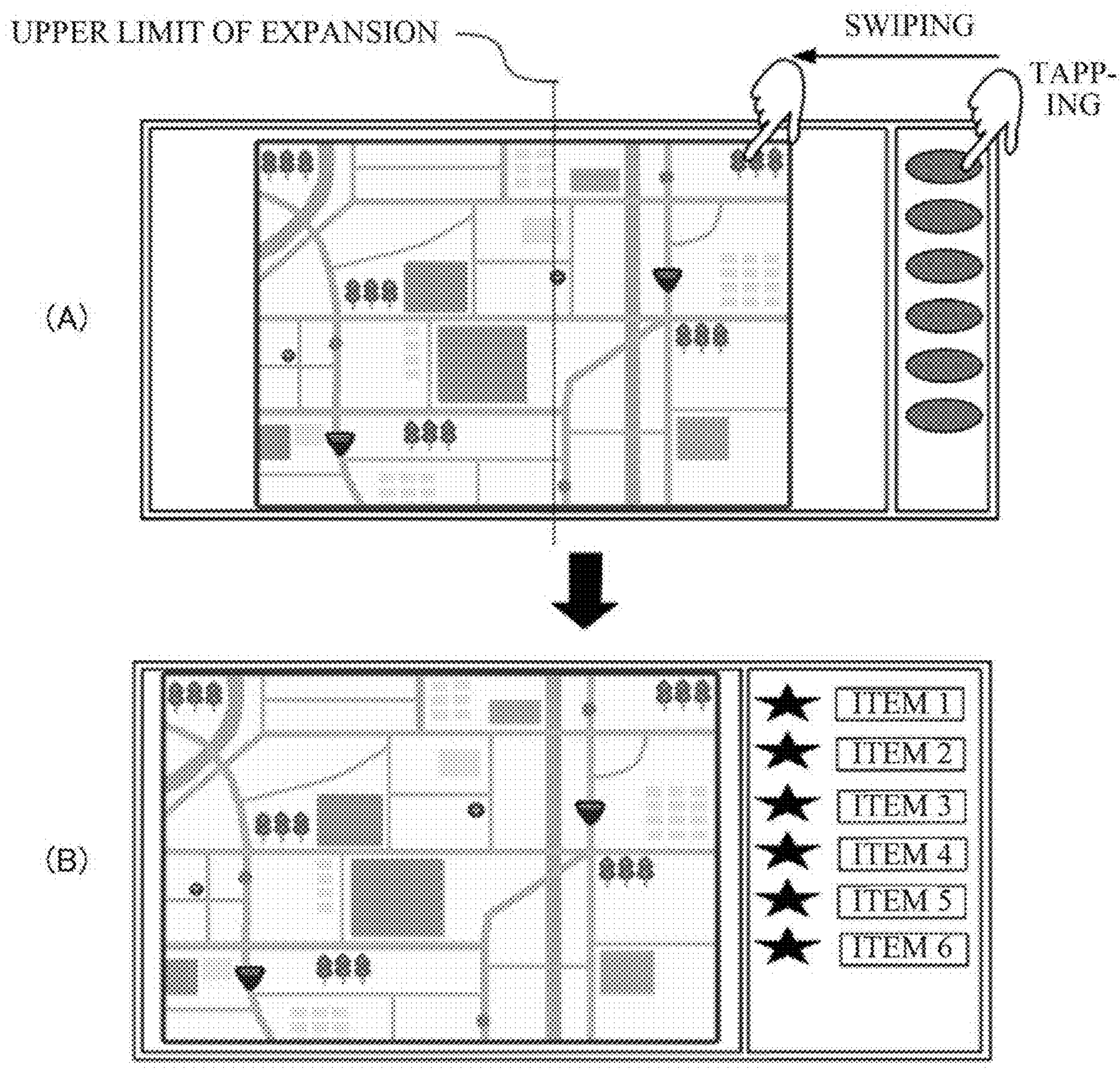

UPPER LIMIT OF EXPANSION
SWIPING
TAPP-
ING
(A)
(B)
ITEM 1
ITEM 2
ITEM 3
ITEM 4
ITEM 5
ITEM 6

FIG. 10

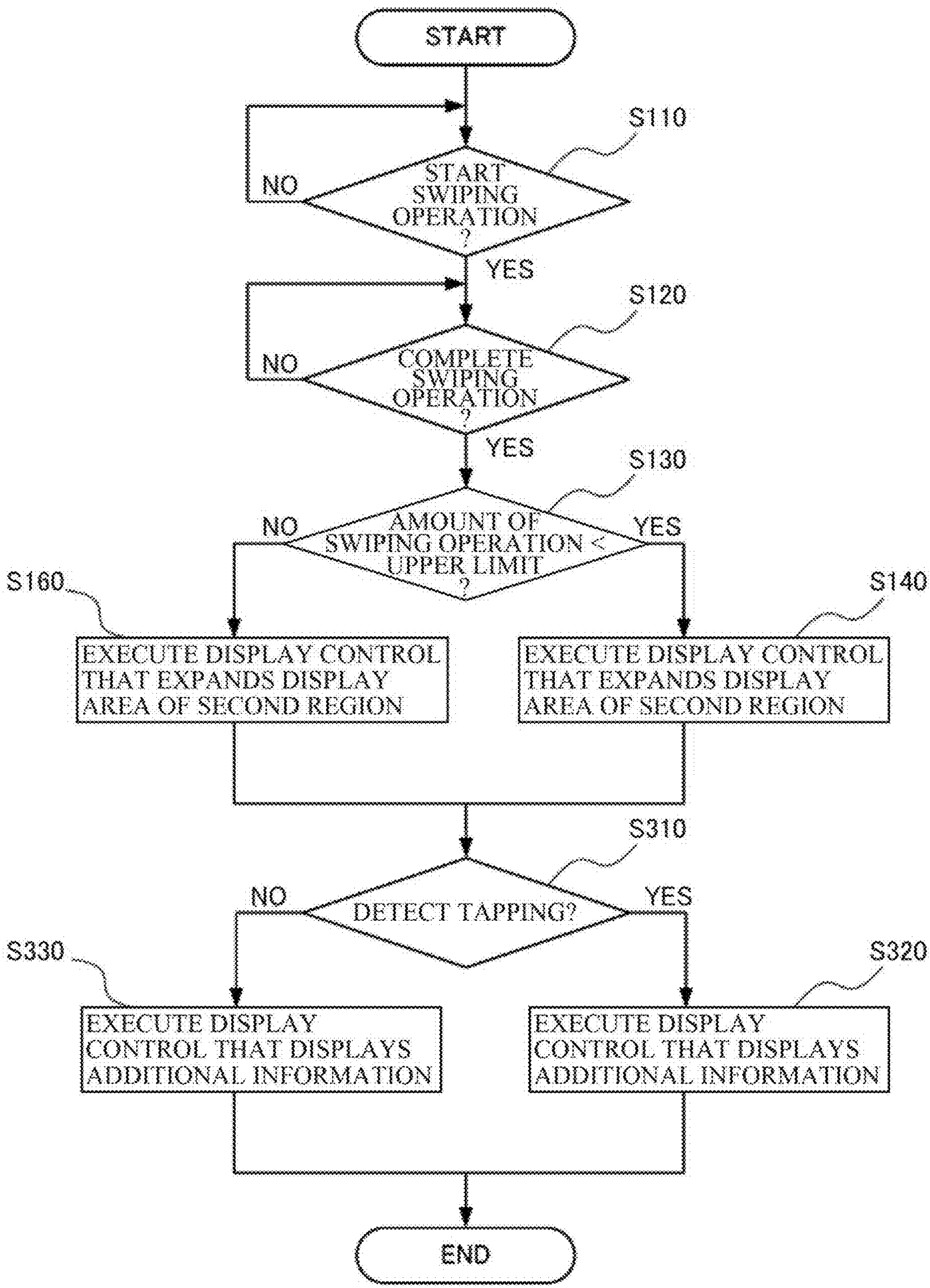

START
S110
NO
START SWIPING OPERATION ?
YES
S120
NO
COMPLETE SWIPING OPERATION ?
YES
S130
NO
AMOUNT OF SWIPING OPERATION < UPPER LIMIT ?
YES
S160
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S140
EXECUTE DISPLAY CONTROL THAT EXPANDS DISPLAY AREA OF SECOND REGION
S310
NO
DETECT TAPPING?
YES
S330
EXECUTE DISPLAY CONTROL THAT DISPLAYS ADDITIONAL INFORMATION
S320
EXECUTE DISPLAY CONTROL THAT DISPLAYS ADDITIONAL INFORMATION
END

FIG. 11

DISPLAY APPARATUS FOR VEHICLE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims priority from Japanese Patent Application No. 2024-033430 filed on Mar. 5, 2024, the entire contents of which are hereby incorporated by reference.

BACKGROUND

[0002] The disclosure relates to a display apparatus for a vehicle.

[0003] A vehicle display apparatus has been widely known that displays various kinds of information including route guidance on a route from a current position of an own vehicle or a current location to a destination, based on both of map data and current position data on the own vehicle received from a positioning satellite system.

[0004] Such a vehicle display apparatus includes, for example, a display having a first region serving as, for example, a main palette, and a second region serving as, for example, a sub-palette.

[0005] For example, a main screen is to be displayed in the first region or main palette, and a shortcut icon to each functionality is to be displayed in the second region or sub-palette.

[0006] When an occupant in the own vehicle selects an icon displayed on the second region or sub-palette, the functionality corresponding to the icon is rendered in the first region or main palette.

[0007] One of the vehicle display apparatuses of this type is disclosed in, for example, Japanese Unexamined Patent Application Publication (JP-A) No. 2010-067129. The vehicle display apparatus includes an image processor, a user's operation detector, and a superimposed image generator. The image processor causes a basic screen including a button image to be displayed on a touch panel. The user's operation detector detects a user's operation made on the button image. When the user's operation is detected, the superimposed image generator generates a pop-up image superimposed on the basic screen to provide a predetermined functionality assigned to the button image. The image processor causes the pop-up image superimposed on the basic screen to be displayed on the touch panel. The pop-up image includes a relation display image that allows a user to visually understand a relation between the button image and the pop-up image. The relation display image is continuously displayed even when display content of the pop-up image is updated.

SUMMARY

[0008] An aspect of the disclosure provides a display apparatus for a vehicle. The display apparatus includes a display, a touch sensor, a display control unit, and a storage. The display includes a first region in which provided content is to be displayed and a second region in which a link icon linking to the content is to be displayed. The display control unit is configured to execute display control that controls a display mode of the display. The storage is configured to store information including data on the display mode of the display. The display control unit is configured to execute the display control to cause a display area of the second region to become smaller than a display area of the first region, and execute, when the touch sensor detects a swiping operation made by an occupant in the vehicle on the link icon displayed on the display, the display control to make expansion of a rendering region of the second region, based on a completion position of the swiping operation made by the occupant, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0009] An aspect of the disclosure provides a display apparatus for a vehicle. The display apparatus includes a display, a touch sensor, one or more processors, and one or or memories. The display includes a first region in which provided content is to be displayed and a second region in which a link icon linking to the content is to be displayed. The one or more processors are configured to execute display control that controls a display mode of the display. The one or more memories are communicably coupled to the one or more processors and configured to store information including data on the display mode of the display. The one or more processors are configured to execute the display control to cause a display area of the second region to become smaller than a display area of the first region, and execute, when the touch sensor detects a swiping operation made by an occupant in the vehicle on the link icon displayed on the display, the display control to make expansion of a rendering region of the second region, based on a completion position of the swiping operation made by the occupant, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The accompanying drawings are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification. The drawings illustrate embodiments and, together with the specification, serve to explain the principles of the disclosure.

[0011] FIG. **1** is a block diagram illustrating a configuration example of a vehicle display apparatus according to one example embodiment of the disclosure.

[0012] FIG. **2** is a diagram illustrating exemplary display modes in first and second regions of a display of the vehicle display apparatus illustrated in FIG. **1**.

[0013] FIG. **3** is a diagram illustrating exemplary display modes in the first and second regions of the display of the vehicle display apparatus illustrated in FIG. **1**.

[0014] FIG. **4** is a flowchart of exemplary processing to be performed by the vehicle display apparatus illustrated in FIG. **1**.

[0015] FIG. **5** is a block diagram illustrating a configuration example of a vehicle display apparatus according to one example embodiment of the disclosure.

[0016] FIG. **6** is a diagram illustrating exemplary display modes in first and second regions of a display of the vehicle display apparatus illustrated in FIG. **5**.

[0017] FIG. **7** is a flowchart of exemplary processing to be performed by the vehicle display apparatus illustrated in FIG. **5**.

[0018] FIG. **8** is a block diagram illustrating a configuration example of a vehicle display apparatus according to one example embodiment of the disclosure.

[0019] FIG. **9** is a diagram illustrating exemplary display modes in first and second regions of a display of the vehicle display apparatus illustrated in FIG. **8**.

[0020] FIG. **10** is a diagram illustrating exemplary display modes in the first and second regions of the display of the vehicle display apparatus illustrated in FIG. **8**.

[0021] FIG. **11** is a flowchart of exemplary processing to be performed by the vehicle display apparatus illustrated in FIG. **8**.

DETAILED DESCRIPTION

[0022] According to a vehicle display apparatus disclosed in JP-A No. 2010-067129, rendering of a functionality in a first region or main palette can be interrupted or suspended when a passenger in a vehicle or a driver who drives the vehicle selects an icon displayed in a second region or sub-palette. Moreover, it is difficult for the passenger or driver to accurately recognize the content of the functionality, simply based on the icon displayed in the second region or sub-palette.

[0023] It is desirable to provide a display apparatus for a vehicle that makes it possible to provide appropriate information displaying to a driver who drives the vehicle by expanding a rendering region of a second region or sub-palette as needed, while securing normal displaying in a first region or main palette.

[0024] In the following, some example embodiments of a vehicle display apparatus according to the disclosure are described in detail with reference to FIGS. **1** to **11**. Note that the following description is directed to illustrative examples of the disclosure and not to be construed as limiting to the disclosure. Factors including, without limitation, numerical values, shapes, materials, components, positions of the components, and how the components are coupled to each other are illustrative only and not to be construed as limiting to the disclosure. Further, elements in the following example embodiments which are not recited in a most-generic independent claim of the disclosure are optional and may be provided on an as-needed basis. The drawings are schematic and are not intended to be drawn to scale. Throughout the present specification and the drawings, elements having substantially the same function and configuration are denoted with the same reference numerals to avoid any redundant description. In addition, elements that are not directly related to any embodiment of the disclosure are unillustrated in the drawings.

First Example Embodiment

[0025] A vehicle display apparatus **1** according to a first example embodiment of the disclosure will now be described with reference to FIGS. **1** to **4**.

<Configuration Example of Vehicle Display Apparatus **1**>

[0026] Referring to FIG. **1**, the vehicle display apparatus **1** according to the first example embodiment includes a processor **100**, a memory **200**, a display **300**, and a touch sensor **400**.

[0027] The processor **100** may execute various kinds of processing that control a display mode of the display **300**, which will be described later, in response to a touch operation made by an occupant of the vehicle (hereinafter simply referred to as "occupant") on the touch sensor **400**, which will be described later.

[0028] In the present example embodiment, the display **300** includes a first region (e.g., main palette) in which provided content is to be displayed, and a second region (e.g., sub-palette) in which link icons linking to the respective pieces of content are to be displayed. The processor **100** executes display control that causes a display area of the second region to become smaller than a display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the processor **100** executes display control that expands a rendering region of the second region, based on a completion position of the swiping operation, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0029] According to the present example embodiment, the processor **100** may further execute display control that causes additional information supplementing the information provided by the link icons to be displayed, in accordance with the expansion of the rendering region, for example.

[0030] Each of the processes will be described in detail later.

[0031] The memory **200** may include a read only memory (ROM) or a random access memory (RAM), and hold and store various programs and various kinds of data.

[0032] The memory **200** may include a storage **210**. The storage **210** may hold and store, for example, the upper limit of expansion of the display area of the second region and display image information on the additional information supplementing the information provided by the link icons.

[0033] The display **300** may display information including image information, illustration information, and text information.

[0034] In some embodiments, the display **300** may be a liquid crystal display.

[0035] The touch sensor **400** may detect contact between the occupant and the display **300**.

[0036] In some embodiments where the display **300** is a liquid crystal display, the touch sensor **400** may be disposed below the display **300**.

[0037] The touch sensor **400** may be any kind of sensing device, such as a resistive touch sensor, a static capacity sensor, an ultrasonic sensor, or an optical (infrared light) sensor.

<Configuration Example of Processor **100**>

[0038] Referring to FIG. **1**, the processor **100** of the vehicle display apparatus **1** according to the present example embodiment may include a display control unit **110** and a processing unit **120**.

[0039] As illustrated in FIG. **1**, devices in the vehicle display apparatus **1** such as the display control unit **110**, the processing unit **120**, the memory **200**, the display **300**, and the touch sensor **400** may be coupled to each other via a bus line BL.

[0040] The display control unit **110** may control a display mode of the display **300**.

[0041] In the present example embodiment, the display **300** includes the first region (e.g., main palette) in which the provided content is to be displayed, and the second region (e.g., sub-palette) in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110** executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110** executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0042] In an example illustrated in FIG. **2**, the occupant may make a swiping operation on any of the link icons displayed on the display **300** illustrated in Part (A) of FIG. **2**, as indicated by an arrow in Part (B) of FIG. **2**. When such a swiping operation is detected, the display control unit **110** may execute the display control that expands the rendering region of the second region, based on a completion position of the swiping operation made by the occupant, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **2**.

[0043] In an example illustrated in FIG. **3**, the occupant may make a swiping operation on any of the link icons displayed on the display **300** illustrated in Part (A) of FIG. **3**, as indicated by an arrow in Part (B) of FIG. **3**. When such a swiping operation is detected, the display control unit **110** may execute the display control that expands the rendering region of the second region, based on a completion position of the swiping operation made by the occupant, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **3**.

[0044] As illustrated in Part (C) of FIG. **2** and Part (C) of FIG. **3**, for example, the display control unit **110** may execute control that causes the additional information supplementing the information provided by the link icons to be displayed, in accordance with the expansion of the rendering region. Non-limiting examples of the additional information may include "setting", "weather", "music", "navigation", "game" and "application".

[0045] The control by the display control unit **110** may be executed based on sensor output data transmitted from the touch sensor **400** to the processing unit **120**, which will be described later, via the bus line BL.

[0046] The processing unit **120** may control an overall operation of the vehicle display apparatus **1**, based on a control program stored in the ROM in the storage **210**, for example.

[0047] In some embodiments, the processing unit **120** may cause the touch sensor **400** to transmit the sensor output data and cause the display control unit **110** to execute the display control.

[0048] In some embodiments, the processing unit **120** may transmit the sensor output data received from the touch sensor **400** to the display control unit **110** via the bus line BL, thereby causing the display control unit **110** to execute the display control.

<Exemplary Processing by Vehicle Display Apparatus **1**>

[0049] Exemplary processing to be performed by the vehicle display apparatus **1** according to the present example embodiment will now be described with reference to FIG. **4**.

[0050] In Step S**110**, the processing unit **120** may determine whether the occupant has started making the swiping operation, based on the sensor output data received from the touch sensor **400**.

[0051] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not started making the swiping operation (Step S**110**: NO), the processing unit **120** may cause the processing to shift to a stand-by mode.

[0052] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has started making the swiping operation (Step S**110**: YES), the processing unit **120** may cause the processing to proceed to Step S**120**.

[0053] In Step S**120**, the processing unit **120** may determine whether the occupant has completed the swiping operation, based on the sensor output data received from the touch sensor **400** (Step S**120**).

[0054] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not completed the swiping operation (Step S**120**: NO), the processing unit **120** may cause the processing to shift to the stand-by mode.

[0055] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has completed the swiping operation (Step S**120**: YES), the processing unit **120** may cause the processing to proceed to Step S**130**.

[0056] In Step S**130**, the processing unit **120** may determine whether the amount of swiping operation made by the occupant is less than a predetermined upper limit, based on the sensor output data received from the touch sensor **400**.

[0057] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is less than the predetermined upper limit (Step S**130**: YES), the processing unit **120** may execute, in Step S**140**, the display control that expands the rendering region of the second region, based on the amount of swiping operation made by the occupant, while continuing to execute the display control in the first region.

[0058] Even after the expansion of the rendering region, the processing unit **120** may continue to execute the display control in the first region, and execute, in Step S**150**, the control that causes the additional information supplementing the information provided by the link icons to be displayed in the second region, in accordance with the expansion of the rendering region. Thereafter, the processing may end.

[0059] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is not less than the predetermined upper limit (Step S**130**: NO), the processing unit **120** may execute, in Step S**160**, the display control that expands the rendering region of the second region, based on the predetermined upper limit, while continuing to execute the display control in the first region. Note that the predetermined upper limit may be set to such a value that the display area of the second region is smaller than the display area of the first region.

[0060] Even after the expansion of the rendering region, the processing unit **120** may continue to execute the display control in the first region, and execute, in Step S**150**, the display control that causes the additional information supplementing the information provided by the link icons to be displayed in the second region, in accordance with the expansion of the rendering region. Thereafter, the processing may end.

<Workings and Effects>

[0061] As described above, the vehicle display apparatus **1** according to the present example embodiment includes the display **300**, the touch sensor **400**, the display control unit **110**, and the storage **210**. The display **300** may display the information including the image information and the text information. The touch sensor **400** may detect contact between the occupant and the display **300**. The display control unit **110** controls the display mode of the display **300**. The storage **210** holds the information including the information on the display mode of the display **300**. The display **300** includes the first region in which the provided content is to be displayed, and the second region in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110** executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects the swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110** executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0062] That is, the display control unit **110** executes the display control that causes the display area of the second region to become smaller than the display area of the first region, and executes, when the touch sensor **400** detects the swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0063] It is therefore possible to provide appropriate information displaying to the driver who drives the vehicle by expanding the rendering region of the second region (e.g., sub-palette) as needed, while securing normal displaying in the first region (e.g., main palette).

[0064] According to a typical technique of changing the size of a display region to a desired size, the size of the display region is changed upon detection of a swiping operation starting from an edge of the display region, based on a completion position of the swiping operation. However, it is difficult for the driver who is driving the vehicle to make such a swiping operation on an edge of the rendering region of the second region (e.g., sub-palette) because the display area of the edge is narrow. In contrast, according to the present example embodiment, the display control unit **110** in the vehicle display apparatus **1** expands the rendering region of the second region upon detection of a swiping operation made by occupant on the link icon on the display **300**, based on the completion position of the swiping operation. This allows the driver who is driving the vehicle to easily change the display size.

[0065] The display control unit **110** in the vehicle display apparatus **1** according to the present example embodiment may cause the additional information supplementing the information provided by the link icons to be displayed, in accordance with the expansion of the rendering region.

[0066] In some embodiments, the display control unit **110** may cause the additional information, such as "setting", "weather", "music", "navigation", "game", and "application", supplementing the information provided by the link icons to be displayed in the second region.

[0067] It is therefore possible to provide appropriate information displaying to the driver who drives the vehicle by expanding the rendering region of the second region (e.g., sub-palette) as needed, while securing normal displaying in the first region (e.g., main palette).

Second Example Embodiment

[0068] A vehicle display apparatus **1**A according to a second example embodiment of the disclosure will now be described with reference to FIGS. **5** to **7**.

<Configuration Example of Vehicle Display Apparatus **1**A>

[0069] Referring to FIG. **5**, the vehicle display apparatus **1**A according to the second example embodiment includes a processor **100**A, a memory **200**A, a display **300**, and a touch sensor **400**.

[0070] Note that components having the same functionalities as those in the first example embodiment are denoted by the same reference numerals to omit detailed description thereof.

[0071] The processor **100**A may execute various kinds of processing that control a display mode of the display **300**, which will be described later, in response to a touch operation made by the occupant of the vehicle on the touch sensor **400**, which will be described later.

[0072] In the present example embodiment, the display **300** includes a first region (e.g., main palette) in which provided content is to be displayed, and a second region (e.g., sub-palette) in which link icons linking to the respective pieces of content are to be displayed. The processor **100**A executes display control that causes a display area of the second region to become smaller than a display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the processor **100**A executes the display control that expands a rendering region of the second region, based on a completion position of the swiping operation, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0073] According to the present example embodiment, the processor **100**A may further execute display control that enlarges the display sizes of the link icons or additional information, in accordance with the expansion of the rendering region, for example.

[0074] Each of the processes will be described in detail later.

[0075] The memory **200**A may include a read only memory (ROM) or a random access memory (RAM), and hold and store various programs and various kinds of data.

[0076] The memory **200**A may include a storage **210**A. The storage **210**A may hold and store, for example, the upper limit of expansion of the display area of the second region, display image information on the additional information supplementing the information provided by the link icons, and display size information on the display sizes of the link icons or additional information.

<Configuration Example of Processor **100**A>

[0077] Referring to FIG. **5**, the processor **100**A of the vehicle display apparatus **1**A according to the present example embodiment may include a display control unit **110**A and a processing unit **120** A.

[0078] As illustrated in FIG. **5**, devices in the vehicle display apparatus **1**A such as the display control unit **110**A, the processing unit **120**A, the memory **200**A, the display **300**, and the touch sensor **400** may be coupled to each other via a bus line BL.

[0079] The display control unit **110**A may control a display mode of the display **300**.

[0080] In the present example embodiment, the display **300** includes the first region (e.g., main palette) in which the provided content is to be displayed, and the second region (e.g., the sub-palette) in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110**A executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110**A executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region in kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0081] In the example illustrated in FIG. **2**, the occupant may make a swiping operation on any of the link icons displayed on the display **300** illustrated in Part (A) of FIG. **2**, as indicated by the arrow in Part (B) of FIG. **2**. When such a swiping operation is detected, the display control unit **110**A may execute the display control that expands the rendering region of the second region, based on a completion position of the swiping operation made by the occupant, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **2**.

[0082] In the example illustrated in FIG. **3**, the occupant may make a swiping operation on any of the link icons on the display **300** illustrated in Part (A) of FIG. **3**, as indicated by the arrow in Part (B) of FIG. **3**. When such a swiping operation is detected, the display control unit **110**A may execute the display control that expands the rendering region of the second region, based on a completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **3**.

[0083] The display control unit **110**A may further execute the display control that enlarges the display sizes of the link icons or additional information in accordance with the expansion of the rendering region, as illustrated in Part (A) to (C) of FIG. **6**, for example.

[0084] The control by the display control unit **110**A may be executed based on sensor output data transmitted from the touch sensor **400** to the processing unit **120**A, which will be described later, via the bus line BL.

[0085] The processing unit **120**A may control an overall operation of the vehicle display apparatus **1**A, based on a control program stored in the ROM in the storage **210**A, for example.

[0086] In some embodiments, the processing unit **120**A may cause the touch sensor **400** to transmit the sensor output data and cause the display control unit **110**A to execute the display control.

[0087] In some embodiments, the processing unit **120**A may transmit the sensor output data received from the touch sensor **400** to the display control unit **110**A via the bus line BL, thereby causing the display control unit **110**A to execute the display control that enlarges the display sizes of the link icons or additional information in accordance with the expansion of the rendering region.

<Exemplary Processing by Vehicle Display Apparatus **1**A>

[0088] Exemplary processing to be performed by the vehicle display apparatus **1**A according to the present example embodiment will now be described with reference to FIG. **7**.

[0089] In Step S**110**, the processing unit **120**A may determine whether the occupant has started making the swiping operation, based on the sensor output data received from the touch sensor **400**.

[0090] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not started making the swiping operation (Step S**110**: NO), the processing unit **120**A may cause the processing to shift to a stand-by mode.

[0091] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has started making the swiping operation (Step S**110**: YES), the processing unit **120**A may cause the processing to proceed to Step S**120**.

[0092] In Step S**120**, the processing unit **120**A may determine whether the occupant has completed the swiping operation, based on the sensor output data received from the touch sensor **400** (Step S**120**).

[0093] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not completed the swiping operation (Step S**120**: NO), the processing unit **120**A may cause the processing to shift to the stand-by mode.

[0094] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has completed the swiping operation (Step S**120**: YES), the processing unit **120**A may cause the processing to proceed to Step S**130**.

[0095] In Step S**130**, the processing unit **120**A may determine whether the amount of swiping operation made by the occupant is less than a predetermined upper limit, based on the sensor output data received from the touch sensor **400**.

[0096] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is less than the predetermined upper limit (Step S**130**: YES), the processing unit **120**A may execute, in Step S**140**, the display control that expands the rendering region of the second region, based on the amount of swiping operation made by the occupant, while continuing to execute the display control in the first region.

[0097] Even after the expansion of the rendering region, the processing unit **120**A may continue to execute the display control in the first region, and execute, in Step S**210**, the display control that causes the additional information supplementing the information provided by the link icon to be displayed and that enlarges the display sizes of the link icons or additional information, in accordance with the expansion of the rendering region. Thereafter, the processing may end.

[0098] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is not less than the predetermined upper limit (Step S**130**: NO), the processing unit **120**A may execute, in Step S**160**, the display control that expands the rendering region of the second region, based on the predetermined upper limit, while continuing to execute the display control in the first region. Note that the predetermined upper limit may be set to such a value that the display area of the second region is kept smaller than the display area of the first region.

[0099] Even after the expansion of the rendering region, the processing unit **120**A may continue to execute the display control in the first region, and execute, in Step S**210**, the display control that causes the additional information supplementing the information provided by the link icons to be displayed and that enlarges the display sizes of the link icons or additional information, in accordance with the expansion of the rendering region. Thereafter, the processing may end.

<Workings and Effects>

[0100] As described above, the vehicle display apparatus **1**A according to the present example embodiment includes the display **300**, the touch sensor **400**, the display control unit **110**A, and the storage **210**A. The display **300** may display the information including the image information and the text information. The touch sensor **400** may detect contact between the occupant and the display **300**. The display control unit **110**A controls the display mode of the display **300**. The storage **210** holds the information including the information on the display mode of the display **300**. The display **300** includes the first region in which the provided content is to be displayed, and the second region in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110**A executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects the swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110**A executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region. Further, the display control unit **110**A may execute the display control that causes the additional information supplementing the information provided by the link icon to be displayed and that enlarges the display sizes of the link icons or additional information, in accordance with the expansion of the rendering region.

[0101] That is, the display control unit **110**A executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects the swiping operation made by the occupant on the link icon displayed on the display **300**, the display control unit **110**A executes the display control that expands the rendering region of the second region, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, based on the completion position of the swiping operation, while continuing to execute the display control in the first region. Further, the display control unit **110**A may execute the display control that causes the additional information supplementing the information provided by the link icon to be displayed, in accordance with the expansion of the rendering region and that enlarges the display sizes of the link icons or additional information.

[0102] It is therefore possible to provide appropriate information displaying to the driver who drives the vehicle by expanding the rendering region of the second region (e.g., sub-palette) as needed, while securing normal displaying in the first region (e.g., main palette), causing the additional information supplementing the information provided by the link icons to be displayed, and enlarging the display sizes of the link icons or additional information.

Third Example Embodiment

[0103] A vehicle display apparatus **1**B according to a third example embodiment of the disclosure will now be described with reference to FIGS. **8** to **11**.

<Configuration Example of Vehicle Display Apparatus **1**B>

[0104] Referring to FIG. **8**, the vehicle display apparatus **1**B according to the third example embodiment includes a processor **100**B, a memory **200**B, a display **300**, and a touch sensor **400**.

[0105] Note that components having the same functionalities as those in the first and second example embodiments are denoted by the same reference numerals to omit detailed description thereof.

[0106] The processor **100**B may execute various kinds of processing that control a display mode of the display **300**, which will be described later, in response to a touch operation made by the occupant of the vehicle on the touch sensor **400**, which will be described later.

[0107] In the present embodiment, the display **300** includes a first region (e.g., main palette) in which provided content is to be displayed, and a second region (e.g., sub-palette) in which link icons linking to the respective pieces of content are to be displayed. The processor **100**B executes display control that causes a display area of the second region to become smaller than a display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the processor **100**B executes the display control that expands a rendering region of the second region, based on a completion position of the swiping operation, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0108] According to the present example embodiment, the processor **100**B may further execute display control that causes a setting screen to be displayed in the second region when a functional setting icon is tapped by the occupant after the expansion of the rendering region, for example.

[0109] Each of the processes will be described in detail later.

[0110] The memory **200**B may include a read only memory (ROM) or a random access memory (RAM), and hold and store various programs and various kinds of data.

[0111] The memory **200**B may include a storage **210**B. The storage **210**B may hold and store, for example, an upper limit of expansion of the display area of the second region, and display information on the setting screen.

<Configuration Example of Processor **100**B>

[0112] Referring to FIG. **8**, the processor **100**B of the vehicle display apparatus **1**B according to the present example embodiment may include a display control unit **110**B and a processing unit **120** B.

[0113] As illustrated in FIG. **8**, devices in the vehicle display apparatus **1**B such as the display control unit **110**B, the processing unit **120**B, the memory **200**B, the display **300**, and the touch sensor **400** may be coupled to each other via a bus line BL.

[0114] The display control unit **110**B may control a display mode of the display **300**.

[0115] In the present example embodiment, the display **300** includes the first region (e.g., main palette) in which the provided content is to be displayed, and the second region (e.g., the sub-palette) in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110**B executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects a swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110**B executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

[0116] In the example illustrated in FIG. **2**, the occupant may make a swiping operation on any of the link icons displayed on the display **300** illustrated in Part (A) of FIG. **2**, as indicated by the arrow in Part (B) of FIG. **2**. When such a swiping operation is detected, the display control unit **110**B may execute the display control that expands the rendering region of the second region, based on a completion position of the swiping operation made by the occupant, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **2**.

[0117] In the example illustrated in FIG. **3**, the occupant may make a swiping operation on any of the link icons displayed on the display **300** illustrated in Part (A) of FIG. **3**, as indicated by the arrow in Part (B) of FIG. **3**. When such a swiping operation is detected, the display control unit **110**B may execute the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region, as illustrated in Part (C) of FIG. **3**.

[0118] In an example illustrated in FIG. **9**, after expanding the rendering region as indicated by an arrow in Part (A) of FIG. **9**, the occupant may tap the functional setting icon, as illustrated in Part (B) of FIG. **9**. When the functional setting icon is tapped by the occupant, the display control unit **110**B may execute the display control that causes the setting screen including "Item **1**" to "Item **6**" to be displayed in the second region, as illustrated in Part (C) of FIG. **9**. In an example illustrated in FIG. **10**, the occupant may tap a setting button at the start of a swiping operation, and then make a swiping operation, as indicated by an arrow in Part (A) of FIG. **10**. When the setting button is tapped by the occupant, the display control unit **110**B may execute the display control that causes the setting screen including "Item **1**" to "Item **6**" to be displayed in the second region, in accordance with the expansion of the rendering region, as illustrated in Part (B) of FIG. **10**.

[0119] The control by the display control unit **110**B may be executed based on sensor output data transmitted from the touch sensor **400** to the processing unit **120**B, which will be described later, via the bus line BL.

[0120] The processing unit **120**B may control an overall operation of the vehicle display apparatus **1**B, based on a control program stored in ROM in the storage **210**B, for example.

[0121] In some embodiments, the processing unit **120**B may cause the touch sensor **400** to transmit the sensor output data and cause the display control unit **110**B to execute the display control. In some embodiments, the processing unit **120**B may transmit the sensor output data received from the touch sensor **400** to the display control unit **110**B via the bus line BL, thereby causing the display control unit **110**B to cause the setting screen to be displayed in the second region when the functional setting icon is tapped by the occupant after the expansion of the rendering region.

[0122] In some embodiments, the processing unit **120**B may transmit the sensor output data received from the touch sensor **400** to the display control unit **110**B via the bus line BL. When the occupant taps the setting button at the start of the swiping operation and then make the swiping operation, the processing unit **120**B may cause the display control unit **110**B to execute the display control that causes the setting screen to be displayed in the second region in accordance with the expansion of the rendering region.

<Exemplary Processing by Vehicle Display Apparatus **1**B>

[0123] Exemplary processing to be performed by the vehicle display apparatus **1**B according to the present example embodiment will now be described with reference to FIG. **11**.

[0124] In Step S**110**, the processing unit **120**B may determine whether the occupant has started making the swiping operation, based on the sensor output data received from the touch sensor **400**.

[0125] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not started making the swiping operation (Step S**110**: NO), the processing unit **120**B may cause the processing to shift to a stand-by mode.

[0126] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has started making the swiping operation (Step S**110**: YES), the processing unit **120**B may cause the processing to proceed to Step S**120**.

[0127] In Step S**120**, the processing unit **120**B may determine whether the occupant has completed the swiping operation, based on the sensor output data received from the touch sensor **400** (Step S**120**).

[0128] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has not completed the swiping operation (Step S**120**: NO), the processing unit **120**B may cause the processing to shift to the stand-by mode.

[0129] If determining, based on the sensor output data received from the touch sensor **400**, that the occupant has completed the swiping operation (Step S**120**: YES), the processing unit **120**B may cause the processing to proceed to Step S**130**.

[0130] In Step S**130**, the processing unit **120**B may determine whether the amount of swiping operation made by the occupant is less than the predetermined upper limit, based on the sensor output data received from the touch sensor **400**.

[0131] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is less than the predetermined upper limit (Step S**130**: YES), the processing unit **120**B may execute, in Step S**140**, the display control that expands the rendering region of the second region, based on the amount of swiping operation made by the occupant, while continuing to execute the display control in the first region.

[0132] After the expansion of the rendering region, the processing unit **120**B may determine in Step S**310** whether the occupant has tapped the functional setting icon.

[0133] If determining that the occupant has tapped the functional setting icon after the expansion of the rendering region (Step S**310**: YES), the processing unit **120**B may cause the display control unit **110**B to display the setting screen in the second region, while continuing to execute the display control in the first region in Step S**320**. Thereafter, the processing may end.

[0134] If determining, based on the sensor output data received from the touch sensor **400**, that the amount of swiping operation made by the occupant is not less than the predetermined upper limit (Step S**130**: NO), the processing unit **120**B may execute, in Step S**160**, the display control that expands the rendering region of the second region, based on the predetermined upper limit, while continuing to execute the display control in the first region. Note that the predetermined upper limit may be set to such a value that the display area of the second region is kept smaller than the display area of the first region.

[0135] If determining that the occupant has not tapped the functional setting icon after the expansion of the rendering region (Step S**310**: NO), the processing unit **120**B may cause, in Step S**330**, the display control unit **110**B to execute the display control that causes the additional information supplementing the information provided by the link icon to be displayed in the second region, in accordance with the expansion of the rendering region, while continuing to execute the display control in the first region even after the expansion of the rendering region. Thereafter, the processing may end.

<Workings and Effects>

[0136] As described above, the vehicle display apparatus **1**B according to the present example embodiment includes the display **300**, the touch sensor **400**, the display control unit **110**B, and the storage **210**B. The display **300** may display the information including the image information and the text information. The touch sensor **400** may detect contact between the occupant and the display **300**. The display control unit **110**B controls the display mode of the display **300**. The storage **210**B holds the information including the information on the display mode of the display **300**. The display **300** includes the first region in which the provided content is to be displayed, and the second region in which the link icons linking to the respective pieces of content are to be displayed. The display control unit **110**B executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects the swiping operation made by the occupant on any of the link icons displayed on the display **300**, the display control unit **110**B executes the display control that expands the rendering region of the second region, based on the completion position of the swiping operation, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region. When the occupant has tapped the functional setting icon after the expansion of the rendering region, the display control unit **110**B may execute the display control that cause the setting screen to be displayed in the second region.

[0137] That is, the display control unit **110**B executes the display control that causes the display area of the second region to become smaller than the display area of the first region. When the touch sensor **400** detects the swiping operation made by the occupant on the link icon displayed on the display **300**, the display control unit **110**B executes the display control that expands the rendering region of the second region, to or below the upper limit of the range where the display area of the second region is kept smaller than the display area of the first region, based on the completion position of the swiping operation, while continuing to execute the display control in the first region. Further, when the occupant has tapped the functional setting icon after the expansion of the rendering region, the display control unit **110**B may execute the display control that causes the setting screen to be displayed in the second region.

[0138] It is therefore possible to provide appropriate information displaying to the driver who drives the vehicle by expanding the rendering region of the second region (e.g., sub-palette) as needed and displaying the setting screen in the second region, while securing normal displaying in the first region (e.g., main palette).

Modification Example 1

[0139] In the first to third example embodiments described above, the display control that expands the rendering region of the second region is executed based on the completion position of the swiping operation made by the occupant. However, according to Modification Example 1, the display control that expands the rendering region of the second region may be executed based on the completion position of the swiping operation made by the occupant, when the occupant makes a tapping operation after the completion of the swiping operation.

[0140] Executing the display control in this manner makes it possible to prevent the occupant from making an erroneous swiping operation.

Modification Example 2

[0141] In the first to third example embodiments described above, the display control that expands the rendering region of the second region is executed based on the completion position of the swiping operation made by the occupant. However, according to Modification Example 2, the rendering region of the second region may be restored to an original size when the amount of swiping operation made by the occupant is less than a predetermined operation amount.

[0142] When the amount of swiping operation made by the occupant is less than the predetermined operation amount, it is difficult to display the additional information supplementing the information provided by the link icon, or although it is possible to display the additional information, the text size can be too small to read.

[0143] To address this concern, according to Modification Example 2, when the amount of swiping operation made by the occupant is less than the predetermined operation amount, the rendering region of the second region may be restored to the original size, and a text message or a sound massage may be issued to urge the occupant to increase the amount of swiping operation.

[0144] In some embodiments, it is possible to implement the vehicle display apparatus **1**, **1**A, or **1**B of any of the example embodiments of the disclosure by recording the process to be executed by the processor **100**, **100**A, or **100**B on a non-transitory recording medium readable by a computer system, and causing the computer system to load the program recorded on the non-transitory recording medium onto the processor **100**, **100**A, or **100**B to execute the program. The computer system as used herein may encompass an operating system (OS) and a hardware such as a peripheral device.

[0145] In addition, when the computer system utilizes a World Wide Web (WWW) system, the "computer system" may encompass a website providing environment (or a website displaying environment). The program may be transmitted from a computer system that contains the program in a storage device or the like to another computer system via a transmission medium or by a carrier wave in a transmission medium. The "transmission medium" that transmits the program may refer to a medium having a capability to transmit data, including a network (e.g., a communication network) such as the Internet and a communication link (e.g., a communication line) such as a telephone line.

[0146] Further, the program may be directed to implement a part of the operation described above. The program may be a so-called differential file (differential program) configured to implement the operation by a combination of a program already recorded on the computer system.

[0147] Although some embodiments of the disclosure have been described in detail with reference to the accompanying drawings, all the vehicle display apparatuses implementable through modifications appropriately made by those skilled in the art based on the vehicle display apparatus according to the foregoing example embodiments of the disclosure also fall within the technical scope of the disclosure as long as they include the gist of the disclosure.

[0148] Within the scope of the idea of the disclosure, those skilled in the art can conceive of various modifications and alterations, and it is understood that these modifications and alterations also fall within the technical scope of the present invention.

[0149] For example, those skilled in the art may add some constituent elements to or delete some constituent elements from the foregoing example embodiments, as appropriate, or add some steps to or remove some steps from the foregoing example embodiment, as appropriate. These embodiments are also included in the technical scope of the disclosure as long as they include the gist of the disclosure.

[0150] Further, it will be understood that other effects than those provided by the foregoing example embodiments and apparent from the description herein or appropriately conceivable by those skilled in the art are naturally provided by the disclosure.

[0151] Various aspects of the disclosure may be provided by appropriately combining the constituent elements disclosed in the foregoing example embodiments.

[0152] For example, some components may be deleted from all the components disclosed in the foregoing example embodiments.

[0153] Furthermore, components across different embodiments may be combined as appropriate.

[0154] The limitations in the claims are to be interpreted broadly based on the language employed in the claims and not limited to examples described in this specification or during the prosecution of the application, and the examples are to be construed as non-exclusive.

[0155] As used in this specification and the appended claims, the singular forms "a", "an", and "the" include, especially in the context of the claims, are to be construed to cover both the singular and the plural, unless otherwise indicated herein or clearly contradicted by context.

[0156] Throughout this specification and the appended claims, unless the context requires otherwise, the terms "comprise", "include", "have", and their variations are to be construed to cover the inclusion of a stated element, integer, or step but not the exclusion of any other non-stated element, integer, or step.

[0157] The use of the terms first, second, etc. do not denote any order or importance, but rather the terms first, second, etc. are used to distinguish one element from another.

[0158] The term "disposed on/provided on/formed on" and its variants having the similar meaning thereto as used herein refer to elements disposed directly in contact with each other or indirectly by having intervening structures therebetween.

[0159] One or both of the display control unit **110** and the processing unit **120** illustrated in FIG. **1** are implementable by circuitry including at least one semiconductor integrated circuit such as at least one processor (e.g., a central processing unit (CPU)), at least one application specific integrated circuit (ASIC), and/or at least one field programmable gate array (FPGA). At least one processor is configurable, by reading instructions from at least one machine readable non-transitory tangible medium, to perform all or a part of functions of the display control unit **110** and the processing unit **120** illustrated in FIG. **1**. Such a medium may take many forms, including, but not limited to, any type of magnetic medium such as a hard disk, any type of optical medium such as a CD and a DVD, any type of semiconductor memory (i.e., semiconductor circuit) such as a volatile memory and a non-volatile memory. The volatile memory may include a DRAM and a SRAM, and the nonvolatile memory may include a ROM and a NVRAM. The ASIC is an integrated circuit (IC) customized to perform, and the

FPGA is an integrated circuit designed to be configured after manufacturing in order to perform, all or a part of the functions of the display control unit **110** and the processing unit **120** illustrated in FIG. **1**.

1. A display apparatus for a vehicle, the display apparatus comprising:

a display comprising a first region in which provided content is to be displayed and a second region in which a link icon linking to the content is to be displayed;

a touch sensor;

a display control unit configured to execute display control that controls a display mode of the display; and

a storage configured to store information comprising data on the display mode of the display, wherein

the display control unit is configured to

execute the display control to cause a display area of the second region to become smaller than a display area of the first region, and

execute, when the touch sensor detects a swiping operation made by an occupant in the vehicle on the link icon displayed on the display, the display control to make expansion of a rendering region of the second region, based on a completion position of the swiping operation made by the occupant, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

2. The display apparatus according to claim **1**, wherein the display control unit is configured to cause additional information supplementing information provided by the link icon to be displayed, in accordance with the expansion of the rendering region of the second region.

3. The display apparatus according to claim **2**, wherein the display control unit is configured to enlarge a display size of the link icon or a display size of the additional information, in accordance with the expansion of the rendering region of the second region.

4. The display apparatus according to claim **2**, wherein

the link icon comprises a functionality setting icon, and

the display control unit is configured to cause a setting screen to be displayed in the second region when the functionality setting icon is tapped by the occupant after the expansion of the rendering region of the second region.

5. A display apparatus for a vehicle, the display apparatus comprising:

a display comprising a first region in which provided content is to be displayed and a second region in which a link icon linking to the content is to be displayed;

a touch sensor;

one or more processors configured to execute display control that controls a display mode of the display; and

one or more memories communicably coupled to the one or more processors and configured to store information comprising data on the display mode of the display, wherein

the one or more processors are configured to

execute the display control to cause a display area of the second region to become smaller than a display area of the first region, and

execute, when the touch sensor detects a swiping operation made by an occupant in the vehicle on the link icon displayed on the display, the display control to make expansion of a rendering region of the second region, based on a completion position of the swiping operation made by the occupant, to or below an upper limit of a range where the display area of the second region is kept smaller than the display area of the first region, while continuing to execute the display control in the first region.

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